Abhinandan

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Experience

Founding Software Engineer Sept 2025 – Oct 2026
Browzer *Remote, San Francisco*

- Built Browzer's Chrome MV3 recorder and CDP-native browser agent with AX/DOM element capture, cross-iframe support, obstruction checks, and real mouse/keyboard/upload execution.
- Reduced LLM cost per automation by ∼**3**× (from avg $0.025 to $0.008 per tool call) across **700K+ monitored production LLM calls** through compact traces, context compression, prompt caching, and dynamic model routing.
- Implemented a streaming ReAct runtime across FastAPI + browser extension with SSE tool execution, multi-tab orchestration, safe parallelism, abort/continue semantics, and auditable traces.
- Shipped zero-LLM replay for intact workflows with stateful AI fallback, allowing failed automations and generated content flows to resume from the point of failure instead of restarting.

Founding Software Engineer Jan 2025 – Jul 2025
Cynos Nexus *Noida, India*

- Built the AI knowledge base service using FastAPI, LangChain, and vector retrieval pipelines, powering customer-facing search workflows.
- Implemented backend infrastructure for retrieval-augmented AI workflows, including document ingestion, structured data processing, asynchronous workers, Redis-backed queues, and production API serving.
- Productionized AI services with Docker, GitHub Actions, Nginx, cloud deployment, asynchronous processing, and API observability.

Contract Software Engineer Sept 2024 – Dec 2024
Etkin.ai *Remote, Turkiye*

- Built a LangChain + pgvector retrieval system; raised recall@5 from 0.71 → 0.82 on a 300-query labeled eval set.
- Developed scalable data-ingestion pipelines using bulk CSV processing and Celery workers, reducing processing time by 40%

Open Source Contributor June 2024 – Aug 2024
HeroUI *Y Combinator S24 startup*

- Received a **personal offer from the CEO** to join HeroUI (prev. NextUI) after **16 merged open-source contributions**.

SWE Intern Jan 2024 – Apr 2024
SkilledUp *Noida*

- Engineered a context-aware AI code assistant for the core learning dashboard using FastAPI and OpenAI.

Open Source Projects

HiQCache | *SGLang, Quantization* | PR #41365 | GitHub

- Implemented INT8 CPU-tier KV caching in SGLang HiCache with GPU↔CPU compression/restoration, reducing Qwen3-8B host KV memory by **43.75%** (144 → 81 KiB/token) and increasing fixed-budget L2 capacity **1.78**× (54K → 96K tokens).
- Benchmarked on an RTX A6000 using exact-token trace replay (128 requests, 284K prompt tokens), improving cache-hit rate from **57.6%** → **72.0%** against BF16 at equal 8 GB budget, capturing ∼**99%** of the workload's 72.72% oracle reuse ceiling.

RolloutCore | *NCCL, vLLM* | GitHub

- Built a RL rollout control plane around vLLM with drain-before-mutate semantics, live NCCL weight installs, post-update cache invalidation, and admit-time version/provenance binding, validated end-to-end without restarting the inference process.
- On Qwen3-8B, measured 4.25s hot updates vs 37.04s full restart (∼8.7× lower model-transition time) on 2× RTX 3090 over PCIe; reproduced 32 stale prefix-cache hits and changed output when cross-version invalidation was deliberately skipped.

MiniServe | *Python, PyTorch* | GitHub

- Built a from-scratch inference runtime with continuous batching, token-budget scheduling, chunked prefill, recompute preemption, and paged KV with content-addressed prefix caching.
- Implemented a Llama-family path over those paged KV tensors and matched Hugging Face (≤6e-8 float32 logit error; identical greedy tokens); on a 120-request shared-prefix workload, prefix reuse cut computed prefill work by 97%.

Technical Skills

ML: PyTorch, Hugging Face, LoRA / PEFT, Qdrant
Inference: vLLM, SGLang, TensorRT-LLM, NCCL, Quantization
Engineering: Python, C++, TypeScript, FastAPI, Node.js, Redis, PostgreSQL, AWS, GCP, Docker, GitHub Actions

Education & Recognition

Dr. APJ Abdul Kalam Technical University B.Tech, Computer Science | 2026

International Youth Math Challenge Gold Honour | **Top 1% TypeScript Engineer Globally** (Algora) | **Amazon ML Summer School 2025**